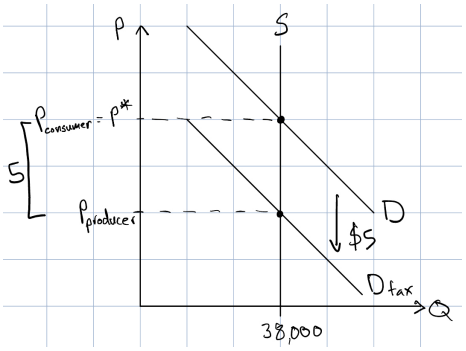
Ch. 6 HW/CW

Principles of Microeconomics

September 30, 2025

1. The tax burden falls entirely on the team owners because their supply of seats is perfectly inelastic. When demand shifts down, the price that consumers pay doesn't move at all, and the price that producers receive goes down by \$5, the full amount of the tax.

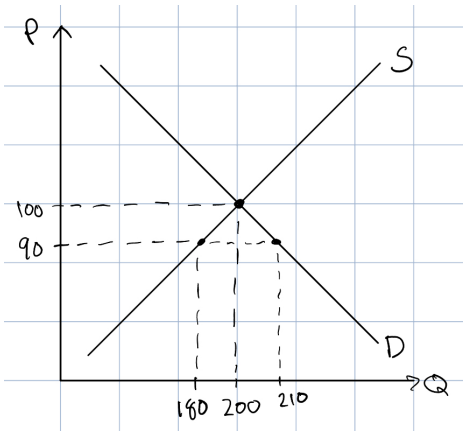


2. (a)

$$\begin{aligned} Q_S &= Q_D \\ 2P^* &= 300 - P^* \\ 3P^* &= 300 \\ P^* &= 100 \\ Q^* &= 2P^* = 2(100) = 200 \end{aligned}$$

(b) A shortage develops because the price ceiling is binding. Therefore, $P = P_{ceiling} = 90$.

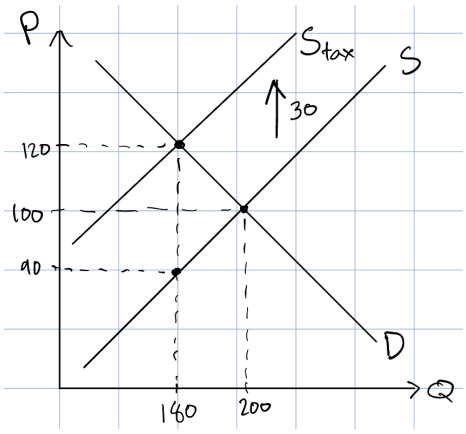
$$\begin{aligned} Q_S &= 2P_{ceiling} = 2(90) = 180 \\ Q_D &= 300 - P_{ceiling} = 300 - 90 = 210 \\ Shortage &= Q_D - Q_S = 210 - 180 = 30 \end{aligned}$$



(c) Neither develops because the price floor isn't binding. Therefore, $P = P^* = 100$, and $Q_S = Q_D = Q^* = 200$.

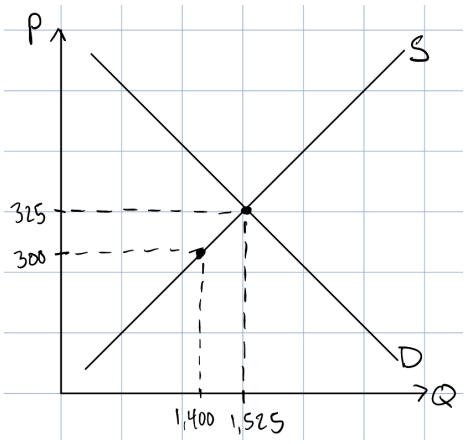
(d) No shortage or surplus develops.

$$
\begin{aligned}
Q_D &= 300 - P_{consumer} \\
Q_S &= 2(P_{consumer} - 30) = 2P_{consumer} - 60 \\
300 - P_{consumer} &= 2P_{consumer} - 60 \\
360 &= 3P_{consumer} \\
120 &= P_{consumer} \\
P_{producer} &= P_{consumer} - 30 = 120 - 30 = 90 \\
Q_{tax} &= Q_S = Q_D = 300 - 120 = 180
\end{aligned}
$$



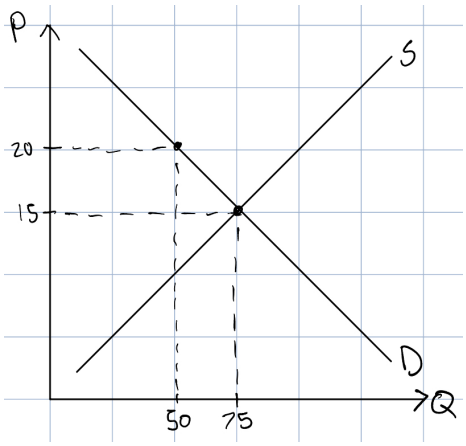
3.

$$
\begin{aligned}
2,500 - 3P^* &= 5P^* - 100 \\
2,600 &= 8P^* \\
325 &= P^* \\
Q^* &= 5(325) - 100 = 1,525 \\
Q_{ceiling} &= 5P_{ceiling} - 100 = 5(300) - 100 = 1,400 \\
\Delta Q &= Q^* - Q_{ceiling} = 1,525 - 1,400 = 125
\end{aligned}
$$



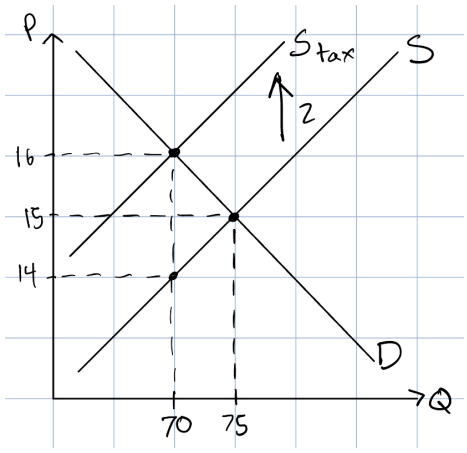
4. (a)

$$
\begin{aligned}
Q_D &= Q_S \\
150 - 5P^* &= 5P^* \\
150 &= 10P^* \\
15 &= P^* \\
Q^* &= 5P^* = 5(15) = 75 \\
Q_{floor} &= 150 - 5P_{floor} = 150 - 5(20) = 50 \\
\Delta Q &= Q^* - Q_{floor} = 75 - 50 = 25
\end{aligned}
$$



(b)

$$
\begin{aligned}
Q_D &= 150 - 5P_{consumer} \\
Q_S &= 5(P_{consumer} - 2) = 5P_{consumer} - 10 \\
150 - 5P_{consumer} &= 5P_{consumer} - 10 \\
160 &= 10P_{consumer} \\
16 &= P_{consumer} \\
P_{producer} &= P_{consumer} - 2 = 16 - 2 = 14 \\
Q_tax &= 150 - 5P_{consumer} = 150 - 5(16) = 70 \Delta Q \quad = 75 - 70 = 5
\end{aligned}
$$



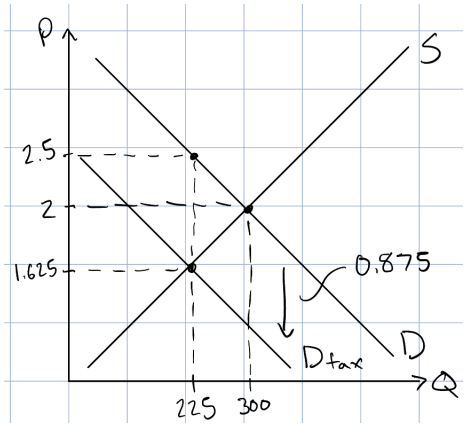
(c) Consumer: $\frac{|P_{consumer} - P^*|}{t} \cdot 100 = \frac{16-15}{2} \cdot 100 = 50\%$
Producer: $\frac{|P_{producer} - P^*|}{t} \cdot 100 = \frac{|14-15|}{2} \cdot 100 = 50\%$

5. (a)

$$\begin{aligned} 600 - 150P^* &= 200P^* - 100 \\ 700 &= 350P^* \\ 2 &= P^* \\ Q^* &= 600 - 150(2) = 300 \end{aligned}$$

(b)

$$\begin{aligned} Q_S &= 200P_{producer} - 100 \\ Q_D &= 600 - 150(P_{producer} + 0.875) = 468.75 - 150P_{producer} \\ 200P_{producer} - 100 &= 468.75 - 150P_{producer} \\ 350P_{producer} &= 568.75 \\ P_{producer} &= 1.625 \\ P_{consumer} &= P_{producer} + 0.875 = 1.625 + 0.875 = 2.50 \end{aligned}$$



(c) $Q_{tax} = 200P_{producer} - 100 = 200(1.625) - 100 = 225$